

# PONMUDI M R

Electronics & Communication Engineering Student | Embedded Systems & RF/SDR Engineer

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## SUMMARY

Second-year ECE student specializing in RF/SDR systems, LoRa mesh networking, and embedded hardware (ESP32/Arduino). Built and led award-winning disaster-communication and GPS-denied positioning systems recognized at national forums (NHIDE 1st Prize, IEEE YESIST12 National Finalist). Looking to bring hands-on embedded/RF product development to a [target role, e.g. Embedded Systems / RF Engineering] team.

## TECHNICAL SKILLS

**Languages:** C, Embedded C, Python

**Frameworks & Platforms:** Arduino, ESP32 (WROOM-32/32D), Zephyr RTOS (STM32)

**RF / SDR Tools:** RTL-SDR V4, SDR++, SDRSharp, SatDump, JAERO, multimon-ng, GNU Radio, fldigi

**Hardware & PCB:** KiCad, PCB Design & Fabrication, LoRa (SX1278), Sensor Integration

**Other Tools:** Git/GitHub, Linux (Fedora), Google Apps Script

## FREELANCE PROJECTS

**TribalNet — Decentralized Disaster Communication & Survivor Tracking** | Team Lead, Team 404 [Jan 2026] – Present

- Delivered a ~10 km line-of-sight disaster-communication network at ~₹1,500 per node by designing a solar-powered ESP32 + SX1278 (433 MHz) LoRa mesh system.
- Earned National Finalist recognition at IEEE YESIST12 2026 for TribalNet's field-ready design.
- Improved survivor-tracking capability by extending the system with RSSI-based triangulation and an ML-driven hazard-prediction module; initiated IPR filing through BIT Sathy's IPR cell.

**Cognitive Opportunistic LEO-PNT Receiver** | iDEX/DRDO Submission, Team 404 [2025] – Present

- Enabled GPS-independent positioning by building an RTL-SDR V4 receiver that extracts Doppler shift from LEO satellites (Orbcomm, NOAA, Iridium).
- Automated signal identification by training a YOLOv8 classifier across 7 RF signal classes on a custom synthetic SDR dataset generator built from scratch.
- Strengthened positioning accuracy under GPS-denied conditions by implementing an EKF/IMU sensor-fusion pipeline.

**Kanaku Book — Payment Ledger PWA** | Freelance Project, SM TEX [month/year] – [month/year]

- Digitized day-to-day ledger tracking for a family textile business by designing and deploying a mobile-first PWA backed by Google Sheets/Apps Script.
- Maintained a clean, auditable contributor history using an orphan-branch git workflow; scoped Play Store distribution via PWABuilder.

**Custom PCB Design & Fabrication** | Independent Project [month/year] – [month/year]

- Took embedded projects from schematic to competition-ready hardware by designing and fabricating custom PCBs in KiCad.
- Enabled multi-platform sensor integration by interfacing ESP32, Arduino Mega, and Raspberry Pi across projects.

## ACHIEVEMENTS & LEADERSHIP

- Winner — 1st Prize (₹50,000) at NHIDE 2026, National Hackathon for Innovation, Design & Entrepreneurship, for TribalNet's Solar-Powered RF Communication design.
- Placed 2nd at Embrix '26 Vegathon (₹10,000), BIT Sathy's national Embedded Systems & Edge AI Hackathon with C-DAC; built the Vega-IoT Gateway on an ARIES v2.0 RISC-V board.
- Placed 2nd at ChronoBIT'26 Hackathon by designing a 555-timer + CD4026 digital clock (HH:MM:SS) with rollover logic via segment decoding.
- Merged an upstream PR to idharshini/Proj-Vega removing hardcoded WiFi/MQTT credentials and fixing a WiFi-init boot hang.
- Google Gemini Student Ambassador and IEEE Electron Devices Society member; active in BIT-IECC and AViNYA Club.

## EDUCATION

**B.E. Electronics and Communication Engineering**

Bannari Amman Institute of Technology (BIT Sathy), Sathyamangalam, Tamil Nadu | 2026 – 2029 (2nd Year)

## LANGUAGES

Tamil (Native), English (Professional), German (Learning — targeting MS in Germany)